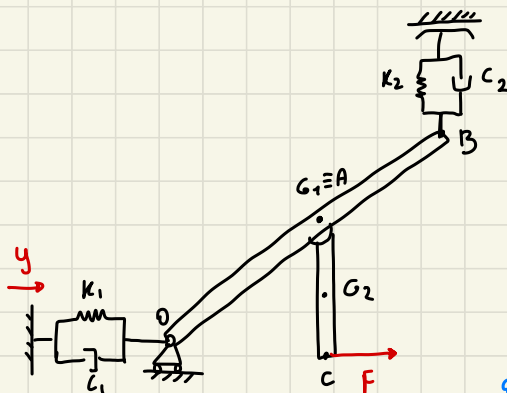


from ESE 5 exercise, continue LAB (ESE 6)



from previously solved problem
we already define $[M]$, $[C]$, $[K]$...
and find the $[M]$, $[C]$, $[K]$

computing ω_n , **FRF**

as I/O Relationship

I: $\cdot F$ @ C applied force } separately

$\cdot y$ constrain displacement } considered
for linearized system with

Superposition principle

overall as sum of two responses

applying just one excitation

considering unitary input, than just if NOT unitary we

scale the response by amplitude

we look the **FRF** (complex value function depending on ω)

depending on Ω forcing frequency assuming

harmonic inputs eg F is stationary periodic with complex

waveform we can still decompose it! $\rightarrow$ deriving total system response

respect **FRF**: function of Ω as excitation pure harmonic

BODE Diagram useful for representation (magnitude + phase) function
 $|FRF|$ $\angle FRF$ of Ω

respect FRF meaning, computation, so on...

RECAP about what is FRF and How we represent it in
complex form $\rightarrow$ physical meaning of $|FRF|$, $\angle$ of the FRF

from the eq. of motion in MATRIX FORM

$$[M]\ddot{x} + [C]\dot{x} + [K]x = \underline{f}(t)$$

$\Rightarrow$ special case $\underline{f} = \text{Re}\{\underline{f}_0 e^{i\omega t}\}$

for a **1DOF** system

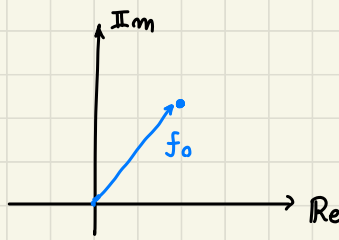
(scalar one harmonic force)

$$\text{Re}\{f_0 e^{i\omega t}\} = f(t)$$

$\hookrightarrow$ GRAPHICALLY

from complex representation

$$f(t) = \text{Re}\{f_0 e^{i\omega t}\}$$



vector representation on Re/Im plane

$f_0 \cdot e^{i\omega t}$ multiplied by $e^{i\omega t}$ complex number for euler's rules

$$e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$$

$$|e^{i\omega t}| = 1 \quad \omega e^{i\omega t} = \omega t$$

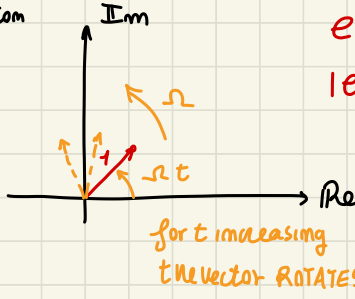
(angle formed by this grow with t through ω)

from the multiplication

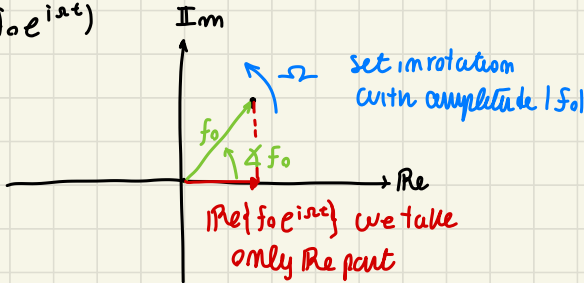
we get

$$|f_0 e^{i\omega t}| = f_0$$

$$\angle f_0 e^{i\omega t} = \angle f_0 + \angle e^{i\omega t} = \angle f_0 + \omega t$$



$$(f_0 e^{i\omega t})$$



if elaborating this DIAGRAM by a Rotation... we can ROTATE by 90° counter clockwise it as:

vector set in rotation with angular speed ω

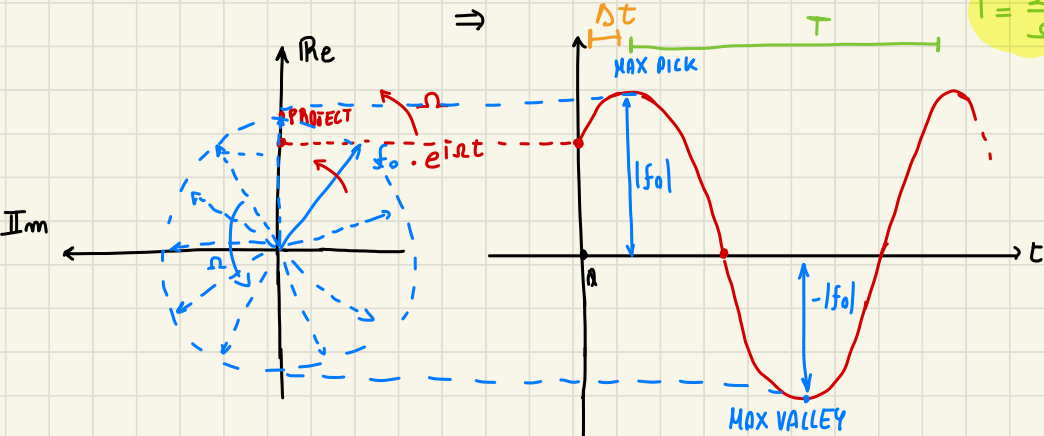
considering $\angle f_0 < 0$

this let us draw the diagram.

$$\text{Re}\{f_0 e^{i\omega t}\}$$

HARMONIC MOTION with PERIOD

$$T = \frac{2\pi}{\omega}$$



So the force $f(t) = \text{Re}\{f_0 e^{i\omega t}\}$ is this representation,
WHICH helps when having same circular frequency you work on PHASOR representation!

for MULTI DOF $\underline{f}(t) = \{\underline{f}_0 e^{i\omega t}\}$ different $|\cdot|$, $\angle \cdot$ $\forall$ terms

So magnitude = excitation amplitude

phase =

because f_0 was $\angle f_0 < 0$ non

zero phase, there is a time delay Δt

relationship of the

complex numbs phase respect time delay of first pick

for just one force, once fixed the $t=0$ is easy

to chose $t=0$ @ Δt , usually we set $\angle f_0 = 0$ by practically

chose the $t=0$ point! SHIFTING time origin

(for a vector multi-phase is NOT possible to have cell in phase study with 0, we should find $|\cdot|$, $\angle \cdot$ per each)

To Represent Input of right hand side of Lagrange

BUT being a LINEAR syst the output will take the form of harmonic waveform with same ω of $\underline{f}(t) \rightarrow$ referring to STATIONARY PART of $\underline{x}(t)$

asympt. stable syst
transient die and
only $\underline{x}_p$ remain

$$\underline{x}_p = \text{Re}\{\underline{x}_0 e^{i\omega t}\}$$

Same form as $\underline{f}(t)$

each $\underline{x}_0$ $|\cdot|$, $\angle \cdot$ represent

amplitude motion of each indep coord
amplitude and delay

So How we relate

$$\underline{f}_0 \rightarrow \underline{x}_0 ?$$

STATIONARY $\rightarrow$ entire I/O information is contained

SIGNALS

on $\underline{f}_0$, $\underline{x}_0$, ω we can draw all signal,
know all!

how to relate $|\underline{f}_0|$, $\angle \underline{f}_0$ TO $|\underline{x}_0|$, $\angle \underline{x}_0$?

$f_0 \rightarrow \underline{x}_0$ for multi DOF, VECTORS...

by forgetting $\text{Re}\{\cdot\}$ we just work in complex domain

$$[m]\ddot{\underline{x}} + [c]\dot{\underline{x}} + [k]\underline{x} = \underline{f}_0 e^{i\Omega t}$$

looking for $\underline{x}_p = \underline{x}_0 e^{i\Omega t}$ (complex solution)

$\begin{cases} \text{Re} \\ \text{Im} \end{cases} \rightarrow \begin{cases} \dot{\underline{x}}_p = i\Omega \underline{x}_0 e^{i\Omega t} \\ \ddot{\underline{x}}_p = -\Omega^2 \underline{x}_0 e^{i\Omega t} \end{cases} \rightarrow \text{plug into equation}$

$$(-\Omega^2[m] + i\Omega[c] + [k])\underline{x}_0 e^{i\Omega t} = \underline{f}_0 e^{i\Omega t}$$

$[A]$ mechanical impedance matrix

$\neq 0$ for $\sin$ $\rightarrow$ eliminating time from eq.

$$\Rightarrow \underline{x}_0 = [A]^{-1} \underline{f}_0$$

(RELATIONSHIP)

$\underline{x}_0 \leftrightarrow \underline{f}_0$ by $[A]^{-1}$ complex matrix

$$[G(i\Omega)] = [A]^{-1}$$

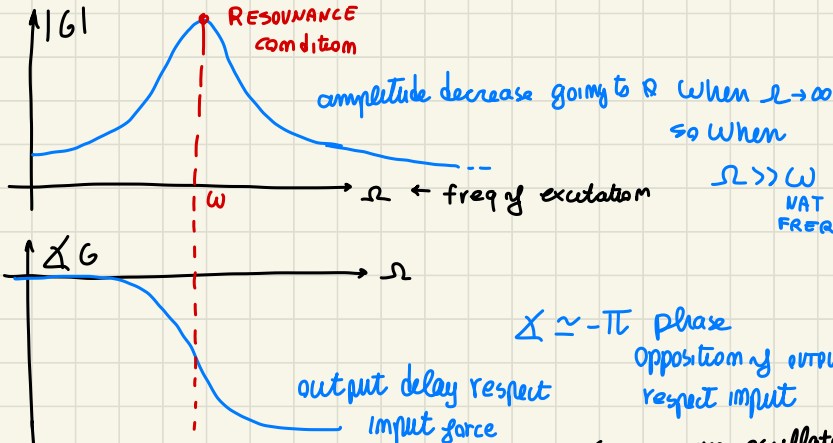
↑
FREQ
TRANS
FUNCT

$$[G(i\Omega)]$$

↳ BODE DIAGRAM

different output depending on Ω

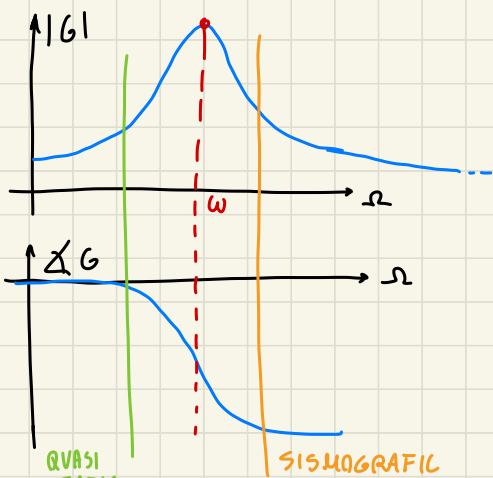
1 DOF
SYSTEM



opposite as expected $\Leftarrow$

(as on an oscillator)
pushing on left it go to the right think dynamically

$\leftarrow \text{move}$

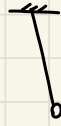


QUASI
STATIC
↓
moving
in same
direction of
excitation

SISMOGRAFIC
↓
moving in
phase opposition

definition
of FRF is
important to
OBTAIN !!

like when moving a PENDULUM



magn/ phase of I/O

are affected by $|G|$, ΔG
relatg I/O relationship



When dealing with this PROBLEM

You will be able to derive FRF

to describe behaviour and design

control syst to affect behav. of syst

(But approaches of control are based on

FRF shaping adding some open

loop control to obtain

$$G_{TOT} = G \cdot G_{ADD} \text{ series}$$

classical
approach



loop control to obtain

MATLAB

as found previously we defined the FRF on a loop

for the two excitations!

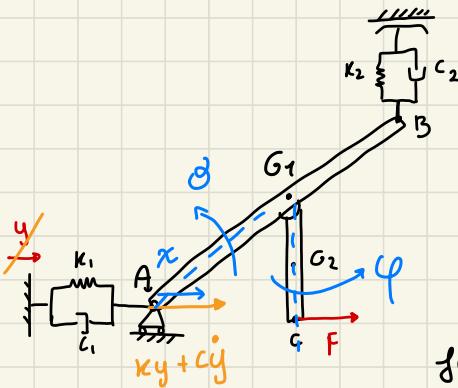
develop 2 SCRIPT parts, for cycle
iteration on $f = [0:0.01:3]$

0 ÷ 3 Hz frequency

compute

FRF in terms of output of x, θ, φ

respect INPUT F OR $K_y + c_y$



for multi DOF we get quantitative result
of BODE plot → 3 RESONANCES pick

the NEXT STEP: for a given INPUT like F KNOWN $\rightarrow$

output may consist of another UNKNOWN $\rightarrow$

FOR EX:

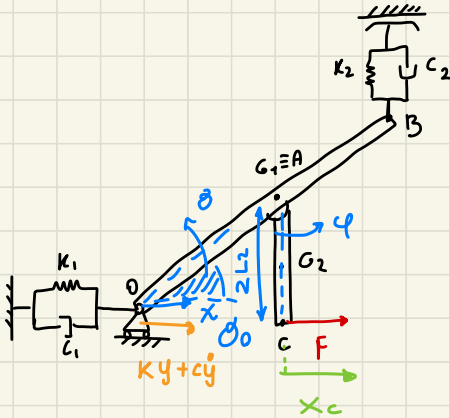
OUTPUT

find FRF representing X_c for a unit input of F

(CO-LOCATED FRF)

take output where we apply input

HOW TO FIND IT?



We use same scheme, as before

apply here UNITARY force unit $F=1$
 $y=0$

$$(-\Omega^2[m] + i\Omega[c] + [k]) \cancel{x_0} e^{i\Omega t} = \frac{1}{f_0} e^{i\Omega t} \quad \text{#0 for shunt}$$

(as done on FIRST for LOOP)

INSTEAD of looking $(x, \theta, \phi)^T$

We wanna look X_c

by relating X_c to our indep. coordinates

Relate $X_c(x, \theta, \phi)$
by KINEMATIC RELAT.
by RIVALS

Looking for linearized

relationship, SMALL x, θ, ϕ around equilibrium

apply same relationship of RIVALS $(\frac{\delta x}{\delta t}) \dots$

$$X_c = X_{G1} + 2L_2\phi = X - L_1 \sin(\theta_0)\theta + 2L_2\phi$$

on POINT of PENDULUM BODY 2

$\rightarrow$ from Kinematics $X_{G1} = X + \dots$

equilibrium FROM JACOBIAN

V_{G1} respect x, θ, ϕ

$\rightarrow$ so from our KNOWLEDGE of the FRF of

x, θ, ϕ respect Ω

Knowing FRF for Input F

and output x, ϕ, θ We can recombine

the FRFs obtain the one with X_c as output from the Relationship found (code)

once coded the FRF script of X_c

we get PICK @ f_1 RESONANCE @ the first natural freq

in fact FIRST mode shape is related to pendulum oscillation,

we don't see other resonance (IF PLOTTING on LOG scale instead you can see other picks very small...)

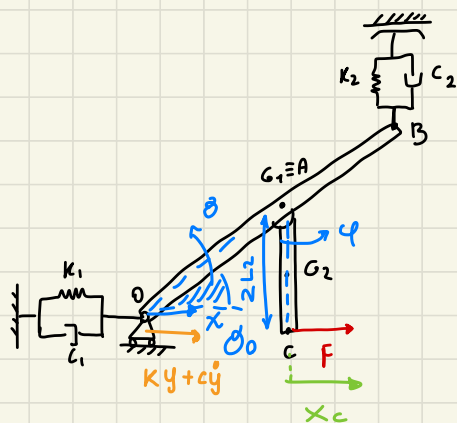
other QUESTION: another interesting OUTPUT

FORCE TRANSMITTED
by SPRING K_2 ?

I can be interested on the FORCE transmitted
by K_2 for example

$$F_{el2} = K_2 \Delta l_2$$

when solving the problem,
we obtain the NON LIN
EXPR. and power series
expansion to II ORD



$$\Delta l_2 = \cancel{\Delta l_{0,2}} - 2L_1 \cos(\theta_0) \cdot \theta + 2L_1 \sin(\theta_0) \theta^2 / 2 + \dots$$

STATIC ELONG. (A) higher order

If F_{el2} as OUTPUT of a LINEAR model

FRF: linear I/O relationship

any terms of order II, III -- neglect!

(linear FRF) so the FRF is I/O relationship

respect on harmonic output magnitude / phase respect dynamic
input harmonic $\rightarrow$ FLUCTUATION around equilibrium (*)

$\Delta l_{0,2}$ related
to STATIC elong.
NOT dym.

We should NOT involve $\Delta l_{0,2}$ which is related to frequencies (STATIC) $\Delta l_{0,2}$ is NOT part of dynamic

elongation $\Rightarrow$ we are interested only on I/O term

so we are interested only on [1/k] 2nd ROW
highlighting only the effect of θ to Δl_2 !

only θ affects Δl_2 related by $-2L_1 \cos(\theta_0)$

so we can define Δl_2 and then $F_{el2} = K_2 \Delta l_2$ PROPORTIONALITY
FORCE transmitted by the spring

IF we are interested on

OVERALL K_2 and C_2 force (parallel spring + dashpot)

$$F_2 = k_2 \Delta l_2 + c_2 \dot{\Delta l_2}$$

time variation
rate of Δl_2

harmonic input and stationary output

$$\Delta l_2 = \Delta l_{2H} e^{i\omega t}$$

(NOT
PRE-LOAD)

harmonic
input
complex value

$$F_2 = k_2 \Delta l_{2H} e^{i\omega t} + c_2 \Delta l_{2H} e^{i\omega t} i \omega =$$

$$= (k_2 + i \omega c_2) \Delta l_{2H} e^{i\omega t}$$

total force
→ SPRING +
DAMPER

→ bring Δl_2 to
an harmonic
complex relation

this can be done
on **MATLAB**

When applying a force on the spring 1, so when $y=1$
excitement on constrain displacement

We see that on F_{el2} BODE plot we get
small effect of resonance of f_1 , in fact does NOT affect a lot the
spring 2
while LARGE peaks @ f_2, f_3 of resonance!

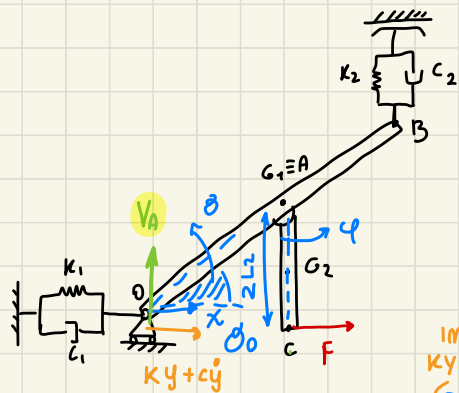
• There is another INTERESTING OUTPUT: CONSTRAINT FORCE

We can evaluate constr force introducing additional displacement
related to y and find from here → NOT ALWAYS BEST approach..

- I can prefer to find constraint force using D'Alembert
Equation!

LAGRANGE usefully want to find all equation → D'Alembert useful when
we want find constrain forces

We wanna find FRF relating
movement of constrain y as INPUT to OUTPUT the vertical constrain
acts on point (A)



D'ALAMBERT to find V :
drawing all the forces acting
(STATIC forces)

static
dynamic
time varying component
 F_{el20} = force damped before spring + damper

• INERTIA terms

ABSOLUTE ROTATION

We can use only dynamical equation or looking only 2 body considering also the forces exchanged internally in between BAR 1/2 if overall terms we neglect internal exchange forces which is canceled for III NEWTON'S LAW

to compute (V_A) : $\sum F_y = 0$:

$$V_A = \underbrace{(m_1 + m_2)g}_{\text{constant static force}} - \overline{F_{el20}} + \underbrace{m_1 \ddot{y}_{G1} + m_2 \ddot{y}_{G2} - F_{el20}}_{V_{ADYN}}$$

That it is NOT considered on PRF
 V_{AST} STATIC TERMS
negligible for I/O relationship

V_{ADYN}
we are interested on it. so we want
• F_{el20} (found previously!)
• $\ddot{y}_{G1}, \ddot{y}_{G2}$ to express

from $y_{G1}(x, \theta, \varphi)$
 $y_{G2}(x, \theta, \varphi)$

seen on V_{G2} respect $\dot{x}, \dot{\theta}, \dot{\varphi}$ JACOBIAN $[1m]$
 V_{G1}

$$y_{G1} = L_1 \cos(\theta_0) \theta \quad (\text{no effect from } x, \varphi)$$

if referring to an harmonic INPUT WITH STATIONARY OUTPUT complex
 $\ddot{y}_{G1} = -\Omega^2 y_{G1}$

@ same way $y_{G2} = L_1 \cos(\theta_0) \theta$

so same expression $\rightarrow$ pendulum motion φ create just $\dot{x}$ vel

$$\ddot{y}_{G2} = -\Omega^2 y_{G2}$$

↳ MATLAB

also Here small effect @ f_1

Big @ f_2, f_3

and here $\ddot{y} \cong \Omega^2$ becomes larger @ resonance higher freq for same displacement...

with $f \uparrow$ larger VIBRATION effect increasing $\ddot{y}_{G1,2}$

different possible TASK of Bode PLOT relating to a given I (excitation/displacement) to a relevant OUT (as an Indep. coord or the displacement of other points of the system) or as OUT we can find an Frel (from measured relationship)

or as OUTPUT a constraining force $\rightarrow$ normally e.g. found by D'ALAMBERT and from complex value phasor extract $| \cdot |, \angle^\circ$ of the force and PLOT IT!